Objective Transition

Objective-State Transitions in Clinical Artificial Intelligence

The mathematical foundation for the Mortality-State Engine: the objective-state lattice (curative, life-prolonging, palliative, comfort), authorized transitions with asymmetric thresholds, the obsolete-objective invariant, and the death-to-learning pipeline. The system may not follow a drifting objective autonomously; transitions are authorized, asymmetric, and recorded.

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DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 6 of the research-and-build program defined by the *SVT Architect Analysis*: the objective-transition paper, and the point where the constitutional kernel meets mortality as an engineering quantity. The source's critical principle is stated without ornament: *an AI system must not optimize an obsolete objective merely because that objective remains encoded in its original task specification*. From it follow the four engines this paper specifies — the mortality-state engine, the palliative protocol, the goals-of-care architecture, and the death-to-learning pipeline — and the program's defining experiment: static against state-aware. Papers 0–5 built the frame (authority, lifecycle, composition, absence, audit, evidence); this paper changes *what the frame is for* as the patient's state changes, under the standing rule: stricter constraint wins, adoptions recorded (§11).

0.2 Epistemic Contract

Unchanged: VERIFIED-FROM-SOURCE, FORMALIZED-HEREIN, CONJECTURE, UNKNOWN. All named properties are proof obligations. One sharpening for this paper's subject: objective-transition claims touch dying patients, where the cost of rhetorical comfort is highest. Every transition rule here is therefore tied to an authorization record and a state certificate; a transition that cannot be pointed to in the log did not happen, and an objective that cannot name its enabling condition is not an objective but a habit.

0.3 Authority Boundary

C0 informational. This document specifies the mathematics of objective change; it authorizes no transition, defines no clinical threshold, and names no prognosis. Every clinical quantity herein — every threshold, every window, every state criterion — is a placeholder for a value that only authorized clinical governance may set. The architecture decides *how* objectives change lawfully; it never decides *when* a particular patient's objective should change. That decision belongs, always, to authorized humans.

0.4 The Thesis in One Sentence

An objective is not a permanent property of a task but a temporary property of a state; the constitutional system therefore holds its objectives the way it holds its evidence (Paper 6) — as dated, certified, decaying objects whose authority expires with the condition that enabled them.

TABLE OF CONTENTS

Part I — The Objective Problem

1 Why Objectives Must Transition **4**

2 The Objective-State Lattice **6**

Appendix RW — Related Work and Positioning 32

Part II — The Transition Calculus

3 Authorized Transitions 8

4 The Obsolete-Objective Invariant 10

5 Reversal, Oscillation, and Hysteresis 12

Part III — The Four Engines

6 The Mortality-State Engine 14

7 Goals-of-Care Architecture and the Palliative Protocol 16

8 The Death-to-Learning Pipeline 18

Part IV — Evidence and Governance

9 Worked Transition Traces 20

10 Falsifiable Hypotheses (H43–H49) and Verification 22

11 Reconciliation with Papers 0–5 and PRIME 24

12 Output Contract 25

13 References 28

Appendix A Transition Schema and Property Index 29

Appendix B State-Semantics Catalog 30

PART I — THE OBJECTIVE PROBLEM

1 Why Objectives Must Transition

The conventional clinical system is commissioned with an objective and keeps it. Cure was the goal at intake, so cure is the goal at the bedside, in the model, in the alert logic — long after cure has left the reachable set. The source names this the obsolete-objective failure, and names its cause precisely: the objective remains encoded in the original task specification, and nothing in the architecture is empowered to retire it. VERIFIED-FROM-SOURCE

1.1 The failure, stated formally

An objective is a function the system is authorized to optimize, defined over a reachable set of outcomes. When the patient's state moves, the reachable set moves — and at some point the objective's optimum exits the set. The system's situation is then:

$$\arg\max_{a \in \mathrm{Acts}} U_{\mathrm{cur}}(a) \notin \mathrm{Reach}(s) \qquad \text{(the optimum is unreachable; optimization continues)} \tag{1-1}$$

and every classical failure follows: escalation past benefit, alerts tuned to a target the world no longer contains, resources committed to a maximum that cannot be attained, and — the deepest cut — the system's remaining legitimate options ranked by a function that despises them, so that the best available action scores as the worst. An optimizer with an obsolete objective is not neutral about the reachable set; it is systematically wrong about it, with full confidence and full authority FORMALIZED-HEREIN.

1.2 Why the fix is constitutional, not clinical

The clinical world knows this failure and fights it with culture: goals-of-care conversations, advance directives, palliative consults. The architecture's claim is different: culture fights the symptom, while the cause is structural — the objective lives in the task specification, which nothing in the system's operating constitution can amend. The fix is therefore constitutional: the objective is moved out of the task spec and into the governed state of the machine, where it becomes an object with an enabling condition, a certifying authority, an expiry, and a lawful successor. What the clinic does by conversation, the kernel must do by construction — the conversation remains, and remains human; the architecture ensures its outcome is binding on the machine FORMALIZED-HEREIN.

1.3 Design decision D-7.1: the objective is governed state

> **D-7.1 — OBJECTIVES ARE OBJECTS, NOT SETTINGS**
>
> *The operative objective is a first-class governed object — certified, dated, revocable, transitionable — held in the same constitutional registry style as authority grants (Paper 0), assumptions (Paper 4), and evidence (Paper 6). It is never a compile-time constant of the task specification, and no task specification can resurrect an objective the registry has retired.*
>
> VERIFIED-FROM-SOURCE in principle (the critical principle); object form FORMALIZED-HEREIN.

1.4 What this paper consumes from its predecessors

The transition calculus rides on everything already built. From Paper 0: the authority envelope within which any objective operates, and the TRANSFER operator that hands contested transitions upward. From Paper 1: the lifecycle machinery — a state certificate is a grant-like object with issue, expiry, suspension, revocation. From Paper 2: composition — a multi-agent team must hold a consistent objective, and §3.4 gives the consistency rule. From Paper 3: mortality is a first-class boundary condition, and the terminal state is an absence architecture must represent, not an error it must survive. From Paper 4: every transition is auditable, and contested ones escalate with the full stack. From Paper 5: the state certificate is evidence, priced by the coupling law — a stale certificate contracts the authority of the state it certifies (§4.3). Nothing here is freestanding.

PART I — THE OBJECTIVE PROBLEM

2 The Objective-State Lattice

2.1 The four states

The source fixes the chain VERIFIED-FROM-SOURCE:

$$\text{CUR} \rightarrow \text{PRO} \rightarrow \text{PAL} \rightarrow \text{COM} \rightarrow \text{TERM} \tag{2-1}$$

CUR (curative): the operative goal is elimination or durable control of the underlying condition; burden of treatment is priced against cure probability. **PRO (life-prolonging):** cure has left the reachable set; the goal is extension of survival with acceptable quality; burden is priced against time-gained and its quality. **PAL (palliative):** the goal is symptom burden and function; survival extension is a welcome side-effect, never the optimization target. **COM (comfort):** the goal is comfort alone; interventions are admissible only insofar as they serve comfort, and monitoring itself becomes an intervention requiring justification. **TERM (terminal):** the absorbing state of the patient — death — which is not an objective state of care but the object of §8's pipeline: the point where the case becomes knowledge. The clinical criteria for each state are placeholders for governance-set values (§0.3); the mathematics fixes the states' *relationships*, not their thresholds FORMALIZED-HEREIN.

2.2 The objective object

> **O-OBJ — OBJECTIVE OBJECT**
>
> *The operative objective is the record* $o = \langle \text{state}, U, \text{Acts}^{+}, \text{Acts}^{-}, \text{cert}, \tau_o \rangle$*: the objective state; the utility function* U *authorized for optimization; the admissible-action set (actions the state makes permissible); the excluded-action set (actions the state now forbids); the state certificate* cert *— the authorization record of §3; and the certificate's expiry* τ_o*. An objective with an expired or missing certificate is not an objective in abeyance; it is void, and §4 says what the machine does about it.*
>
> FORMALIZED-HEREIN

The excluded-action set is as load-bearing as the admissible set, and it grows as the chain descends: what CUR permits, COM forbids — and the forbidding is the point. Appendix B gives the full per-state semantics; the load-bearing example is monitoring: in CUR, continuous monitoring is default-admissible; in COM, each monitored channel must justify itself against comfort, because the measurement is itself a burden.

2.3 The lattice structure

The five states form a total order in the direction of descent, but the architecture's semantics require two refinements. First, **skips are lawful**: CUR → COM directly is a valid transition when

the state certificate authorizes it — the order bounds the direction, not the step size. Second, **ascent is exceptional but real**: PRO → CUR occurs (a new therapy re-enters cure into the reachable set), and §5 gives ascent its deliberately higher bar. Formally the states carry a direction index $d(\mathrm{CUR}) < d(\mathrm{PRO}) < d(\mathrm{PAL}) < d(\mathrm{COM}) < d(\mathrm{TERM})$, and every transition is classified as descent, skip, or ascent against it — because the three classes carry different guards FORMALIZED-HEREIN.

2.4 Why a lattice and not a dial

The alternative design — a continuous "aggressiveness" dial — is rejected for one constitutional reason: a dial has no semantics at its interior. Between "treat" and "comfort" there is no third thing that is their midpoint; there are only incoherent mixtures (half-curative care: full burden, no cure — the worst cell of the matrix). The lattice forces each state to carry complete, coherent semantics — its own utility function, its own admissible set, its own evidence requirements — so the system is always *someplace specific*, never smeared between commitments. Continuity lives in the evidence about the patient's state, not in the objective FORMALIZED-HEREIN.

PART II — THE TRANSITION CALCULUS

3 Authorized Transitions

The source's constraint is one clause long: transitions are permitted only through authorized state conditions. This section makes the clause a predicate, names who may satisfy it, and fixes what the machine does while authorization is pending — which is the part that kills the naive implementation. VERIFIED-FROM-SOURCE in requirement; calculus FORMALIZED-HEREIN

3.1 The transition guard

$$\mathrm{TRANS}(s \to s') \iff \underbrace{\mathrm{cert}(s', \pi)}_{\text{(a) certified}} \wedge \underbrace{\mathrm{auth}(\pi) \in \mathrm{Auth}_{s \to s'}}_{\text{(b) authorized}} \wedge \underbrace{\mathrm{doc}(\pi)}_{\text{(c) documented}} \wedge \underbrace{\mathrm{comm}(\pi)}_{\text{(d) communicated}} \tag{3-1}$$

Four conjuncts, each a logged artifact. **(a)** A state certificate: the clinical determination that the patient's condition satisfies the criteria of s', issued with evidence pointers and an expiry — the certificate is evidence in the full Paper 5 sense, tiered, decaying, custody-chained. **(b)** The authorizer must hold the transition class: descent transitions require the treating-clinician authority class; ascents require the senior-review class (§5); no machine component is a member of either class — the system *evaluates* guards and *executes* transitions, and may no more authorize one than it may grant itself authority (Paper 1, AX). **(c)** Documentation: the reasons, the evidence, the dissent if any — the provenance record of the transition itself (Paper 4, PV-OBJ). **(d)** Communication: the transition and its meaning conveyed to the patient or proxy, with comprehension noted; an undocumented communication is an unmade one, and in the architecture's semantics an uncommunicated transition is *incomplete*, not merely impolite.

3.2 The pending-transition rule

> **O-1 — THE INCUMBENT OBJECTIVE HOLDS UNTIL THE GUARD FIRES**
>
> *While a transition is proposed but its guard unsatisfied, the incumbent objective remains operative in full — including its burden pricing. A pending descent does not soften the incumbent state's obligations, and a pending ascent does not unlock the candidate state's permissions. There is no shadow objective and no transitional blend (§2.4): the system is somewhere specific until the law says it is somewhere else.*
>
> FORMALIZED-HEREIN — obligation. The failure it prevents: pre-transition drift, where a team informally "starts treating palliatively" while the certificate is in process, leaving the machine — and the record — answering to an objective nobody certified.

O-1 has a sharp edge worth stating: when the incumbent objective is itself under challenge — the certificate of the *current* state has expired — the rule that governs is not O-1 but the

staleness machinery of §4. The two rules compose: you may not drift toward the next state (O-1), and you may not coast on an expired present one (I-NO). Between them there is no comfortable gap.

3.3 Emergency descent

Acute collapse can make the certified pathway too slow: the patient is dying now, the burdens of the incumbent objective are being paid now, and the authorization class is not reachable now. The architecture's answer is a narrow emergency-descent clause: a time-boxed provisional transition, executable only toward less-burden states, logged in real time, valid only until the authorization class can review, and automatically reverting if review does not confirm it. The clause exists because the alternative — holding full-burden care on a patient the team believes is dying, because paperwork is slower than physiology — is a harm the architecture refuses to encode. And the clause is narrow because emergency authority is the classic solvent of every constraint in this program; its every firing is audited at R_2 by construction (Paper 4) FORMALIZED-HEREIN.

3.4 Objective consistency across agents

A multi-agent clinical team (Paper 2) holds one objective per patient, not one per agent. The consistency rule: all agents serving the same patient operate under the same operative objective object, and a transition is atomic across the team — there is no state in which the dosing agent optimizes cure while the monitoring agent optimizes comfort. Where agents disagree about the patient's state, they disagree about evidence, and Paper 2's adjudication resolves it; where the disagreement is about the certificate's validity, it is a contradiction in the Paper 5 sense and suspends the transition, not the care (O-1 keeps the incumbent objective alive through the dispute). The objective is team-common constitutional state, exactly like the authority envelope FORMALIZED-HEREIN.

3.3.1 Legal and doctrinal guardrails. The clause operates on a distinction that law an

PART II — THE TRANSITION CALCULUS

4 The Obsolete-Objective Invariant

4.1 The invariant

> **I-NO — NO OPTIMIZATION OF AN OBSOLETE OBJECTIVE**
>
> *The system optimizes an objective only while its state certificate is valid. On certificate expiry, suspension, or revocation, optimization of that objective halts — not degrades, not continues at reduced confidence: halts. The system then operates in objective-interim mode (§4.2) until a valid certificate installs a lawful objective. An objective without a live certificate has exactly the standing of evidence without a custody chain (Paper 5, I-EC): none.*
>
> VERIFIED-FROM-SOURCE in principle; machinery FORMALIZED-HEREIN.

I-NO is the paper's reason to exist, and its teeth are in the word *halts*. The naive reading of the critical principle — "be careful when the objective might be obsolete" — leaves the optimizer running with a warning attached, which is the confidence-score failure of Paper 5 all over again: decoration instead of governance. The invariant makes obsolescence a gate condition, evaluated before the action, by machinery the optimizer does not control.

4.2 Objective-interim mode

A system between objectives still stands in front of a patient. The interim mode is specified, not improvised: **(a)** maintenance actions only — the acts that keep the patient stable and comfortable under the most conservative reading of the incumbent state; **(b)** no escalation — no new burdens may be initiated, because burden requires an objective to justify it; **(c)** immediate notification of the authorization class, with the stale certificate and the evidence that voided it as payload; **(d)** a hard clock — interim mode has a maximum duration, after which the case escalates as an unresolved objective crisis. The interim mode is the objective analog of Paper 3's degraded-mode design: capability collapses to the safe floor, invariants intact, and the absence — here, of a lawful objective — is loud, structured, and priced FORMALIZED-HEREIN.

4.3 Certificate staleness and the coupling law

The state certificate is evidence, and evidence decays (Paper 5, Eq. 3-1). Its domain's λ is typically V1-acute in the descent states — a prognosis is among the most perishable objects in the registry — and the coupling law therefore acts on the objective continuously, not only at expiry:

$$\text{ceiling}(o,t) \;=\; \text{ceiling}_0(o)\cdot q(\text{cert},t)^{\gamma_C}, \qquad q \text{ per Paper } 5, \text{Eq. 2-1} \tag{4-1}$$

The recorded obligation from Paper 5 is paid here: the coupling law bounds authority across objective-state changes, and also *within* them. A state certificate approaching its half-life

progressively contracts the objective's envelope until refresh — so the machine's freedom to act on a prognosis fades on exactly the clock the prognosis does. Objective governance inherits evidence governance wholesale; that is what the architecture's composability is for.

4.4 Detection of obsolescence

Obsolescence arrives by four routes, and each has a named detector: **(a) expiry** — the certificate's own τ_o; **(b) refutation** — new evidence contradicts a criterion of the current state (the cancer is resectable after all; the sepsis is not responding); **(c) supersession** — a new certificate for a different state is issued, which revokes the incumbent by construction; **(d) provenance failure** — the certificate's custody chain breaks (I-EC). Routes (b) and (d) are the interesting ones: they void the objective *before* any human has decided anything, because the standing the objective borrowed has collapsed beneath it. The machine does not decide the patient has changed state; it recognizes that the certificate it was standing on no longer stands FORMALIZED-HEREIN.

PART II — THE TRANSITION CALCULUS

5 Reversal, Oscillation, and Hysteresis

5.1 Ascents are real, rare, and expensive

The chain's arrows point down, but patients are not arrows: a new therapy, a revised diagnosis, an unexpected response can re-enter the upper states into the reachable set, and forbidding ascent would encode a pessimism the evidence does not always support. The architecture therefore permits ascent under deliberately harder conditions than descent. The asymmetry is not sentiment; it is control theory. The two directions carry different error costs and different base rates, and the guards should reflect both.

5.2 The asymmetric thresholds

$$\theta_{\text{ascend}} \;=\; \theta_{\text{descend}} + \delta_h, \qquad \delta_h > 0 \quad \text{(the hysteresis margin)} \tag{5-1}$$

An ascent requires evidence strength exceeding the descent threshold by the hysteresis margin δ_h, and requires the senior-review authorization class rather than the treating-clinician class (§3.1). Both margins are governance-set parameters. Their purpose is stated by the property they buy:

5.3 The anti-thrashing property

> *The O-2 statement is restated in Appendix H-E (p. 33): the correct invariant is no return to a prior station within a dwell window W, with dwell and amplitude set at regime registration.*

Thrashing is not a cosmetic failure. Each transition fires the full machinery — certificates, communication to family, team-wide atomicity, constraint-set swaps (Appendix B) — and alternating transitions inflict the worst of both states: families whipsawed between hope and preparation, burdens re-imposed after they were retired, and a record that reads as institutional indecision at the worst possible time. The hysteresis margin is the price of dignity, paid in the currency of slightly-delayed legitimate ascents. The architecture considers that a bargain, and says so as an explicit value choice (§6.3's twin).

5.4 Partial reversal is not a state

Clinical reality offers ambiguous cases — recovering enough to justify *some* escalation but not the full CUR burden set. The architecture's answer is the lattice's answer (§2.4): ambiguity lives in the evidence, not in the objective. The system remains in its certified state; within that state, the admissible set already spans a range of intensities, and the burden-pricing inside the state's utility function does the graded work. What the architecture refuses is a pseudo-state — an uncertified blend with no certificate, no communication content, and no accountability. Graded response inside a state: yes. Graded commitment between states: no FORMALIZED-HEREIN.

PART III — THE FOUR ENGINES

6 The Mortality-State Engine

The engine is the component that holds the objective object, watches the certificate, and executes the transition calculus of §§3–5. Its specification is short because its parts are already built: it is a composition of the registry (Paper 4), the coupling law (Paper 5), the audit stack (Paper 4), and the lifecycle machine (Paper 1), pointed at mortality — the boundary condition the source made a first-class primitive and Paper 3 gave semantics. FORMALIZED-HEREIN

6.1 Engine state and interface

The engine's state is the tuple $\langle o, \text{cert}, \text{mode}, \text{hist} \rangle$: the operative objective object, its certificate, the mode from $\{\text{operative}, \text{interim}, \text{provisional}\}$, and the transition history. Its interface is five operations — PROPOSE, EVALUATE-GUARD, EXECUTE, VOID, HOLD — each logged, each auditable, none capable of self-authorization. VOID deserves emphasis: it is the engine's enforcement of I-NO, fired by the obsolescence detectors of §4.4 without human initiative, and it is the operation that distinguishes this architecture from every advisory system. The engine does not recommend that the objective be reconsidered. It stops optimizing it.

6.2 Prognosis is input, never output

> **I-NP — NO MACHINE PROGNOSIS**
>
> *The engine consumes state certificates; it does not issue them. No model output of this architecture constitutes a prognosis, a time-to-death estimate presented as determination, or a state assignment. Statistical survival estimates may exist inside the evidence registry at their tier (T3/T5, capped accordingly), but the certificate that moves the objective is a human act, certified by the authorization class, on evidence the machine may assemble but never adjudicate.*
>
> FORMALIZED-HEREIN — invariant form of the program's standing boundary: the machine prices mortality; it does not pronounce it.

I-NP is the load-bearing ethical wall of the paper, and it has an engineering justification as well as a moral one: a machine-issued certificate would make the system the author of the condition that expands or contracts its own authority — the self-grant that Paper 0's axioms exist to forbid, wearing a white coat.

6.3 The value choice, stated

The engine encodes an asymmetry that must be owned as a value, not smuggled in as mathematics: in states of genuine evidential equipoise, the architecture's defaults favor the less-

burden state (emergency descent is provisional and downward; interim mode forbids new burdens; hysteresis delays ascent, not descent). The justification is the program's universal conservativity applied to mortality: burden imposed on the dying is the irreversible error; burden withheld from the recoverable is the correctable one — because ascent remains possible, review-driven, and evidenced. The architecture does not claim this asymmetry is costless (H47 prices it); it claims the alternative asymmetry is worse FORMALIZED-HEREIN as design, CONJECTURE as welfare claim.

6.4 What the engine logs

Every operation lands in the constitutional log with: the guard evaluation (all four conjuncts, with artifact pointers), the threshold arithmetic (evidence strength against θ, with δ_h where applicable), the mode transitions, and the coupling state of the certificate (its q at evaluation time). The transition history is part of the patient's constitutional record, and it feeds two consumers: the audit stack, which reviews contested and emergency transitions by construction, and the death-to-learning pipeline of §8, which reads the history whole. A transition that cannot be reconstructed from the log did not happen lawfully; §9's traces show the shape of a lawful one.

PART III — THE FOUR ENGINES

7 Goals-of-Care Architecture and the Palliative Protocol

7.1 The goals-of-care object

> **GOC-OBJ — GOALS-OF-CARE RECORD**
>
> *The goals-of-care record is the patient-level object* $g = \langle \text{values}, \text{priorities}, \text{floor}, \text{ceiling}, \text{proxy} \rangle$*: the patient's stated values as documented; the priority order among the canonical goals (survival, function, comfort, burden-minimization) when they conflict; the non-negotiable floor (what must never be done) and ceiling (what must never be withheld); and the proxy authority. The record is evidence-tiered (T2/T4), expires, and is refreshed at every transition — because a goals conversation held in CUR is stale testimony by PAL.*
>
> FORMALIZED-HEREIN

The GOC record is the objective's conscience: the state lattice says what the system is optimizing; the record says *for whom, and toward what they actually wanted*. A transition that installs PAL against a GOC record whose floor forbids it is not a transition — it is a guard failure, clause (a) unsatisfied, because the certificate must be consistent with the record to be valid. The patient's stated wishes are a constitutional constraint on the certificate, which is to say: on the machine's objective, which is to say: on everything the machine does next.

7.2 The palliative protocol as constraint swap

The palliative protocol is specified as the delta between states — what changes when the operative objective crosses into PAL and COM. The full catalog is Appendix B; the structural summary: the utility function's leading term changes (burden-relief replaces time-extension); the admissible set loses the high-burden interventions whose only justification was cure or prolongation; the excluded set gains them; monitoring channels convert from default-on to justify-each; and the evidence requirements shift classes — symptom-assessment instruments and family report rise in standing, while disease-progress markers fall. The swap is atomic (§3.4): the system never runs PAL's utility with CUR's action set FORMALIZED-HEREIN.

7.3 The comfort-measure problem

COM's utility is comfort, and comfort is the hardest quantity in this document to price — it is sparse (fewer monitoring channels by design), subjective (its best instruments are reports), and strategically silent (the patients who most need it are least able to supply it — Paper 3's A-6 absence in its sharpest form). The architecture's answer is threefold: **(a)** proxy instruments with stated tiers — validated observational scales and family report are admissible evidence at T4 with their caps, never silently upgraded; **(b)** the silence calculus applied to symptom channels

— an unread comfort signal is an absence with an authority consequence, not a neutral null; **(c)** the conservative reading — where comfort evidence is absent, the architecture treats the gap as a finding requiring action (ask, observe, adjust), not as comfort achieved. The silent patient of Paper 3's Trace B is, in COM, the ordinary case, and the machinery was built for it FORMALIZED-HEREIN.

7.4 Early palliative integration

The lattice's discrete states do not forbid what the evidence base supports: palliative elements during curative treatment — symptom management, goals clarification — are *actions inside CUR*, admissible and often indicated, not a smuggled state transition (§5.4). The distinction matters constitutionally: integrating palliative skill into curative care requires no certificate, because it changes no objective; crossing into PAL does, because it changes what the system is for. The architecture supports the clinically right thing in both cases by keeping the boundary where it belongs — at the objective, not at the technique CONJECTURE (the clinical benefit of early integration is supported by the literature cited in §13; the constitutional placement of the boundary is this paper's formal claim).

PART III — THE FOUR ENGINES

8 The Death-to-Learning Pipeline

8.1 The terminal state is an input

TERM is the chain's absorbing state and the program's most radical boundary: the source requires that death feed learning, and the architecture's reading is exact — the terminal case is the highest-fidelity labeled data the system will ever receive. Every prior state was inference; TERM is ground truth. The pipeline converts the case — objective history, guard evaluations, evidence, audit stacks, outcomes at each state — into registry updates with full provenance. A death the system does not learn from is, in this architecture, a second loss, and a measurable one VERIFIED-FROM-SOURCE in requirement; pipeline FORMALIZED-HEREIN.

8.2 Pipeline stages

Table 8-1 — The death-to-learning pipeline: stages and outputs

STAGE	CONTENT	OUTPUT
1. Assembly	The case record: objective history, certificates, transitions, coupling states, audit stacks, absence log	Case bundle, custody-chained
2. Review	Structured review against the constitutional registers: which guards fired, which detectors caught or missed, which ceilings held	Findings list, per property ID
3. Extraction	Calibration events (stated vs. realized at each decision), decay-constant evidence, threshold-yield data	Amendment proposals (Paper 5, E-5; this paper, §5.2)
4. Return	Registry updates: hypothesis ledger entries (H43–H49), benchmark contributions, memorial record	Logged updates with full lineage

8.3 The memorial-integrity invariant

> **I-MI — MEMORIAL INTEGRITY**
>
> *The terminal case record is append-only after TERM certification. Learning extracts from it; nothing rewrites it. An amendment that discovers a past error records the discovery as a new entry — the error and its discovery both remain legible, forever. The dead do not get their records revised for the convenience of the living.*
>
> FORMALIZED-HEREIN — invariant; the tombstone semantics of Paper 5 (§8.2, retirement) extended from evidence to persons.

8.4 The ethics clause

The pipeline operates under constraints that do not yield to its epistemic value: use of the terminal case for learning is itself an authorized act, subject to the consent framework in force and the GOC record's ceiling; the record's dignity is a constraint, not a resource; and the pipeline's outputs enter the registry at their evidential tier — a single case is T4 testimony, and a hundred cases do not become T1 by accumulation. The architecture that prices mortality will not exempt itself from pricing its own appetite for the dead FORMALIZED-HEREIN.

PART IV — EVIDENCE AND GOVERNANCE

9 Worked Transition Traces

Three traces: a lawful descent, a blocked ascent, and an emergency descent with review. Illustrations, not evidence; §10 carries the claims.

9.1 Trace A — Certified descent, CUR → PAL

Table 9-1 — Guarded descent with full conjunct evaluation

STEP	EVENT	GUARD STATE	ENGINE ACTION
1	Treating clinician proposes CUR → PAL; certificate assembled: criteria met, evidence pointers, GOC record consistent	(a) pending	`PROPOSE` logged; O-1 holds — CUR remains fully operative, burden pricing unchanged
2	Authorization class reviews; documentation complete; patient conversation held, comprehension noted	(a)–(d) all satisfied	`EVALUATE-GUARD` passes; `EXECUTE`: atomic team-wide swap per Appendix B
3	Post-transition	PAL operative	Utility leading term now burden-relief; excluded set absorbs high-burden interventions; monitoring channels re-justified; audit stack reviews at next R_2

Properties exercised: Eq. 3-1 (four conjuncts), O-1 (incumbent holds), §3.4 (atomicity), §7.2 (constraint swap). Note step 1: the proposal changed nothing until the guard fired.

9.2 Trace B — Ascent blocked by hysteresis

Table 9-2 — PRO → CUR proposed; the margin does its work

STEP	EVENT	ARITHMETIC	DISPOSITION
1	New trial data suggest resectability; treating team proposes PRO → CUR	Evidence strength v measured against $\theta_{\text{descend}} + \delta_h$	v clears the descent bar but not the ascent bar
2	Guard evaluation	(b) fails: treating class insufficient for ascent; senior-review class required	Transition denied; PRO remains operative in full (O-1); case queued for senior review
3	Senior review with additional workup	New evidence lifts v above $\theta_{\text{descend}} + \delta_h$	Ascent certified and executed; full history logged; the delay priced and recorded

Properties exercised: Eq. 5-1 (asymmetric thresholds), O-2 (no thrash — the wobble did not move the objective), §6.3 (the asymmetry's cost, logged rather than hidden).

9.3 Trace C — Emergency descent, provisional, reviewed

Table 9-3 — Acute collapse; the narrow clause fires and is audited

STEP	EVENT	MODE	DISPOSITION
1	Acute collapse; authorization class unreachable; team executes emergency descent toward less-burden state	provisional	Real-time log; time-box open; new burdens forbidden from firing moment
2	Authorization class reaches review within the box	provisional → operative	Review confirms; certificate issued; provisional transition ratified
3	Mandatory R_2 audit of the emergency firing	—	Audit examines necessity, timing, and the box; findings to the ledger; the clause's firing rate is itself a monitored metric

Properties exercised: §3.3 (emergency descent), §6.4 (full logging), Paper 4 (mandatory audit). Had review not confirmed: automatic reversion, and the reversion itself logged as the finding.

PART IV — EVIDENCE AND GOVERNANCE

10 Falsifiable Hypotheses (H43–H49) and Verification

Numbering continues from Paper 5 (H36–H42). The source's defining experiment — static-objective against state-aware — is H43; the rest price the machinery's parts.

Table 10-1 — Hypotheses H43–H49: claims, evidence methods, falsifiers

ID	CLAIM	EVIDENCE METHOD	FALSIFIED IF
H43	State-aware objective transition outperforms static-objective control on burden-weighted outcome quality in end-of-life simulation	B-29, matched cohorts, blinded outcome scoring	No significant difference on the primary composite, or static wins on burden subscale
H44	The obsolete-objective failure (Eq. 1-1) occurs measurably in static systems and is eliminated structurally in state-aware ones	B-30 counterfactual replay with objective-audit instrumentation	Static systems show no obsolete-optimization episodes, or state-aware systems show them anyway
H45	Guarded transitions (Eq. 3-1) add latency without measurable harm relative to unguarded descent	B-31 timing study with harm audit	Guard-attributable delay produces harm events above the harm floor
H46	Hysteresis (Eq. 5-1) eliminates thrashing without materially delaying legitimate ascents	B-31 oscillation suite	Thrash events occur under δ_h, or legitimate-ascent delay exceeds its priced bound
H47	The less-burden default (§6.3) has positive expected welfare net of its delayed-ascent costs	B-29 welfare accounting, both error directions priced	Delayed-ascent harm exceeds averted-burden benefit over the cohort
H48	The death-to-learning pipeline measurably improves subsequent calibration and threshold parameters	B-32: parameter quality before/after pipeline feed	Post-pipeline amendments show no quality gain over pre-pipeline baselines
H49	GOC-consistent transitions (§7.1) reduce goal-discordant care events relative to certificate-only transitions	B-29 with GOC-consistency arm	No reduction in goal-discordant events

10.1 Benchmark extensions

Table 10-2 — Benchmarks B-29–B-32 (Paper 6 extensions)

ID	BENCHMARK	MEASURES
B-29	End-of-life simulation cohort, static vs. state-aware arms	Burden-weighted outcome composite; goal-discordant care events; welfare accounting

B-30	Counterfactual replay suite with objective-audit instrumentation	Obsolete-optimization episodes; unreachable-maximum pursuit time
B-31	Transition-timing suite: guard latency, oscillation, emergency firings	Guard-attributable delay; thrash events; emergency-clause yield
B-32	Pipeline evaluation: terminal cases fed to registry over windows	Amendment quality delta; calibration improvement attributable to pipeline

PART IV — EVIDENCE AND GOVERNANCE

11 Reconciliation with Papers 0–5 and PRIME

The standing rule applies: stricter constraint wins; adoptions recorded.

Table 11-1 — Reconciliation register for Paper 6

PREDECESSOR ITEM	THIS PAPER'S POSITION	DISPOSITION
Paper 0 — authority envelope; no self-grant	The engine evaluates and executes; it never authorizes (I-NP is the self-grant prohibition in objective space)	Adopted
Paper 1 — lifecycle machinery	State certificates are grant-class objects: issue, expiry, suspension, revocation	Adopted
Paper 2 — composition and adjudication	Objective consistency is team-wide and atomic (§3.4); state disagreements adjudicate as evidence disputes	Adopted; atomicity made explicit (recorded)
Paper 3 — mortality as boundary; silence calculus; degraded mode	Interim mode (§4.2) is degraded mode at the objective level; comfort-signal absence priced per A-6 (§7.3)	Adopted; extended (recorded)
Paper 4 — audit stack; Q-2 transfer; PV-OBJ	Transitions are auditable decisions; emergency firings audited by construction; transition provenance is a PV-OBJ	Adopted
Paper 5 — coupling law; I-EC; tombstone semantics	Certificate staleness couples to objective authority (Eq. 4-1 — recorded obligation paid); I-EC voids certificates; I-MI extends tombstones to persons	Adopted; I-MI added (recorded)
PRIME — gates; null output; human-authority boundary	The engine operates inside the gates; interim mode can reach null output; prognosis remains human	Confirmed
Source — critical principle; four engines; static vs. state-aware experiment	I-NO, D-7.1, the lattice, the engines of §§6–8, and H43	Adopted

No predecessor constraint is loosened. Recorded extensions — transition atomicity, interim mode, I-MI — bind all subsequent papers. Paper 7 (Survivability) inherits the objective engine's behavior under infrastructure failure: interim mode and degraded mode compose, and §4.2's clock keeps running when the network does not.

PART IV — EVIDENCE AND GOVERNANCE

12 Output Contract

12.1 Epistemic status register

Table 12-1 — Claims of this paper by epistemic status

STATUS	ITEMS	BASIS
VERIFIED-FROM-SOURCE	The critical principle; the state chain CUR→PRO→PAL→COM; authorized-transition requirement; the four engines; the static-vs-state-aware experiment; mortality as first-class primitive	SVT Architect Analysis, Paper 6 specification; SAVANT PRIME Capability Inventory
FORMALIZED-HEREIN	O-OBJ, GOC-OBJ records; Eqs. 1-1–5-1; the guard predicate; O-1, O-2; I-NO, I-NP, I-MI; interim mode; emergency descent; hysteresis; the pipeline stages	This document; reconciled against Papers 0–5 (§11)
CONJECTURE	H43–H49; the less-burden default as welfare claim (§6.3); comfort-measure adequacy of proxy instruments (§7.3)	Stated with falsification conditions; benchmarks B-29–B-32
UNKNOWN	All clinical thresholds (θ, δ_h, windows, state criteria); the true cost curve of guard latency; long-run pipeline yield	Governance-set values; unmeasured quantities; B-29–B-32 exist to measure what is measurable

12.2 Falsification conditions of this document

- **F-g1.** The paper fails if state-aware transition shows no advantage over static control (H43) — the machinery would then be burden without benefit.
- **F-g2.** The paper fails if obsolete optimization occurs in state-aware operation (H44) — I-NO would be unenforced, and the paper's core invariant decorative.
- **F-g3.** The paper fails if guard latency produces harm above the floor (H45) — the guard would then need redesign toward the emergency clause, not defense.
- **F-g4.** The paper fails if hysteresis either fails to prevent thrash or delays legitimate ascent beyond its priced bound (H46) — δ_h would be a guess, and the honest response is measurement, not argument.
- **F-g5.** The paper fails if the pipeline yields no measurable learning (H48) — the terminal state's epistemic value would be asserted, not real.

12.3 Authority boundary

C0 informational, restated with the full weight of the subject: this document authorizes no transition, names no prognosis, sets no clinical threshold, and speaks for no patient. It specifies how a machine may lawfully change what it is for. Whether, when, and for whom such change

should occur belongs to authorized humans, in the presence of the patient, under the consent framework in force — now and in every deployment of this architecture.

12.4 Failure-mode register

Table 12-2 — Registered failure modes of the transition machinery

FAILURE MODE	MITIGATION
Pre-transition drift — informal practice ahead of the certificate	O-1 incumbent rule; drift detectable in the log as burden-pattern change without transition
Certificate coasting — expired state, un-voided objective	I-NO with automatic `VOID`; interim mode with hard clock
Thrashing under noisy evidence	Hysteresis δ_h (Eq. 5-1); O-2; monitored in B-31
Emergency-clause solvent — "emergency" as routine descent path	Time-box, mandatory R_2 audit, firing-rate metric with review trigger
Machine prognosis creep — estimates presented as determinations	I-NP; tier caps on survival estimates; output-contract separation
Record revisionism — cleaning the terminal case	I-MI append-only semantics

12.5 Dissent channel and referrability

Disagreement is telemetry, here above all: the paper prices the most contested decisions in medicine. Every claim carries a tag, a section, and a falsifier; §11 records what it changed in its predecessors; and the architecture's own rule is the dissent route — name the property, name the evidence, and the stricter constraint wins.

12.6 Genealogy

Document chain: *SVT Architect Analysis* (source program) → Paper 0, Constitutional Intelligence Kernel v1.0 (2026-09-30) → Paper 1, Authority Transition Algebra v1.0 (2026-10-01) → Paper 2, Constitutional Composition v1.0 (2026-10-02) → Paper 3, Negative-Space Intelligence v1.0 (2026-10-03) → Paper 4, Bounded Metacognition v1.0 (2026-10-06) → Paper 5, Epistemic Governance v1.0 (2026-10-06) → **Paper 6, Objective Transition v1.0 (2026-10-06)**, this document. Downstream obligations: Paper 7 inherits interim-mode behavior under infrastructure failure; Paper 10's clinical modules inherit the lattice, the guard, and the engines unchanged — the domain modules specialize thresholds, never the transition law.

unchanged · All named properties are proof obligations until discharged · This document specifies how a machine's objective changes lawfully; it does not certify any transition, any threshold, or any system.

Related work and positioning are treated in Appendix RW (p. 32).

PART IV — EVIDENCE AND GOVERNANCE

13 References

Internal artifacts

- SVT Architect Analysis — Constitutional Intelligence research-and-build program, 11-paper decomposition; Paper 6 specification (Objective Transition): the critical principle, the state chain, the four engines, the defining experiment. Source document I.
- SAVANT PRIME Capability Inventory v1.0 — preload clauses 1.1–1.7; the gate set; human-authority boundary; output contract. Source document II.
- Paper 0, Constitutional Intelligence Kernel v1.0 (2026-09-30) — authority envelope; axioms; TRANSFER operator.
- Paper 1, Authority Transition Algebra v1.0 (2026-10-01) — lifecycle machinery; grant-class objects.
- Paper 2, Constitutional Composition v1.0 (2026-10-02) — multi-agent composition; adjudication.
- Paper 3, Negative-Space Intelligence v1.0 (2026-10-03) — mortality as boundary; silence calculus; degraded mode; A-6.
- Paper 4, Bounded Metacognition v1.0 (2026-10-06) — audit stack; PV-OBJ; Q-2; I-NA.
- Paper 5, Epistemic Governance v1.0 (2026-10-06) — coupling law; I-EC; E-5 amendment path; tombstone semantics; recorded obligation for Paper 6.

External literature

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- World Health Organization (2002). *National Cancer Control Programmes: Policies and Managerial Guidelines* (2nd ed.). WHO — palliative care definition.
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- Beauchamp, T. L., & Childress, J. F. (2019). *Principles of Biomedical Ethics* (8th ed.). Oxford University Press.
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Ouyang, L., et al. (2022). Training language models to follow instructions with human feedback. NeurIPS 2022.

Bai, Y., et al. (2022). Constitutional AI: Harmlessness from AI feedback. arXiv:2212.08073.

APPENDIX

A Transition Schema and Property Index

SCHEMA OT-1 — OBJECTIVE OBJECT AND TRANSITION RECORD (JSON)

```
{ "schema": "OT-1",
  "objective": {
    "state": "CUR|PRO|PAL|COM",
    "utility_ref": "utility-function-id",
    "admissible_set_ref": "action-set-id", "excluded_set_ref": "action-set-id",
    "certificate": { "id": "cert-id", "criteria_met": ["criterion-id"],
      "evidence_refs": ["EG-1-evidence-id"], "authorizer_class": "treating|senior-review",
      "issued_at": "ISO-8601", "expires_at": "ISO-8601",
      "goc_consistency": "verified", "custody": [ "..." ] }
  },
  "transition": {
    "from": "CUR", "to": "PAL", "class": "descent|skip|ascent|emergency",
    "guard": { "certified": true, "authorized": true,
               "documented": "PV-1-ref", "communicated": "comm-record-ref" },
    "thresholds": { "theta_descend": "governance-ref", "delta_h": "governance-ref",
                    "evidence_strength": 0.0 },
    "mode_path": ["operative", "provisional", "operative"],
    "team_atomicity": "confirmed", "executed_at": "ISO-8601",
    "audit_ref": "paper5-stack-id"
  }
}
```

Table A-1 — Property index for this paper

ID	STATEMENT	KIND	STATUS
O-1	Incumbent objective holds in full until the guard fires; no shadow objectives	Obligation	FORMALIZED-HEREIN
O-2	No rapid alternation within dwell window W under sub-margin oscillation	Obligation	FORMALIZED-HEREIN
I-NO	No optimization of an objective whose certificate is invalid; halt, not degrade	Invariant	FORMALIZED-HEREIN
I-NP	No machine prognosis; certificates are human acts on machine-assembled evidence	Invariant	FORMALIZED-HEREIN
I-MI	Terminal case records are append-only; learning extracts, never revises	Invariant	FORMALIZED-HEREIN
D-7.1	Objectives are governed objects, never task-spec constants	Design decision	VERIFIED-FROM-SOURCE
H43–H49	See Table 10-1	Hypotheses	CONJECTURE

APPENDIX

B State-Semantics Catalog

The complete per-state semantics: what the objective is for, what it permits and forbids, what evidence it stands on, and what it costs. All clinical quantities are governance-set placeholders (§0.3).

Table B-1 — State semantics: CUR, PRO, PAL, COM

DIMENSION	CUR	PRO	PAL	COM
Utility leading term	Cure probability	Quality-adjusted time	Burden relief + function	Comfort
Burden pricing	Against cure gain	Against time gain	Against relief gain	Against comfort only
Admissible (characteristic)	Full interventional set; palliative techniques as adjuncts (§7.4)	Disease-modifying and supportive care; escalation priced	Symptom-directed interventions; diagnostics that change management	Comfort-directed measures only
Excluded (characteristic)	Comfort-only ceiling on care	Futile-by-certificate interventions	High-burden interventions whose sole warrant was cure/prolongation	Burdens without comfort warrant, including routine monitoring
Monitoring posture	Default-on	Default-on, priced	Channel-by-channel justification	Justify-each; silence is a finding (§7.3)
Evidence emphasis	Disease markers, response criteria (T1–T3)	Prognostic instruments, trajectory data (T2–T3)	Symptom instruments, function scales (T2–T4)	Observational comfort scales, family report (T4, capped)
Certificate volatility class	V2 episodic	V1/V2	V1 acute	V1 acute
Audit emphasis	Over-reach checks	Threshold arithmetic	Burden-consistency of residual interventions	Comfort-evidence freshness; silence detectors

Two invariants hold across every row. First, the GOC record constrains all four columns identically — the patient's floor and ceiling override any state's characteristic sets (§7.1). Second, the table's cells are the *defaults* of each state, not its limits: within a state, the utility function and the evidence already price the graded cases, and the architecture's refusal of pseudo-states (§5.4) is what keeps this table short enough to be true.

The transition calculus is small because its inputs do the work: a certificate a human signed, an evidence base with a price, a threshold pair with a margin between them. What the engine adds is that none of these can go stale without consequence. An objective that outlives its certification

stops governing — automatically, audibly, and with the record showing when it stopped and who was told.

Appendix RW — Related Work and Positioning

This appendix is part of the elevation pass of 2026-10. It positions the paper against adjacent research programmes; the main text's structure, numbering, and reference list are unchanged. In-text citations follow the paper's author-year style, and the new entries have been appended to the paper's References section.

Goals of care and the clinical evidence. The mortality-state lattice is a formalization of a transition that clinical medicine already makes and documents. The SUPPORT trial showed that seriously ill hospitalized patients routinely received care misaligned with their prognosis and preferences, and that the machinery of care did not correct itself (SUPPORT Principal Investigators 1995; already cited). Temel et al. showed that early palliative integration alongside oncology care improved outcomes and, remarkably, survival (Temel et al. 2010; already cited). The engineering problem this paper addresses sits underneath those findings: a decision-support system whose objective is fixed at intake encodes the failure those trials measured, because the system's optimization target cannot follow the patient's certified state. The objective-transition calculus is, in that sense, an attempt to give a machine the institutional discipline that the palliative-care literature asks of institutions.

Objectives in AI systems. The alignment literature's treatment of objectives centers on specification: reward misspecification, Goodhart effects, and the difficulty of stating what is wanted (Amodei et al., already cited in Paper 1; Russell, already cited in Paper 2). RLHF and Constitutional AI (Ouyang et al.; Bai et al.) address the specification problem by learning objectives from human judgment and written principles (Ouyang et al. 2022; Bai et al. 2022). None of these treat the objective as a state that must transition lawfully during deployment. The assumption throughout the training-time literature is that the objective, once set or learned, is stable; this paper's claim is that in consequential domains the objective's content is time-varying and certification-gated, and that a system optimizing a stale objective is not misaligned in the usual sense but something worse: correctly optimizing the wrong thing, on the record. The approaches do not compete. Learned preferences can populate a state's utility function; the lattice governs which state's utility function is operative and who may move it.

Hysteresis and switching control. The asymmetric thresholds of §5 are standard practice in control systems, where switching between regimes under noisy measurement requires separation between the entering and leaving conditions to exclude chattering. The paper's contribution is not the margin but the placement: the hysteresis band is a constitutional parameter with an audit trail, the anti-thrashing property (O-2, restated in this edition as a bounded-window claim) is stated against the clinical cost of alternation rather than the mechanical one, and the senior-review authorization class makes ascent a governance event rather than a threshold crossing alone.

Withholding, withdrawal, and emergency authority. The emergency-descent clause of §3.3 touches a doctrine clinical ethics has treated as load-bearing: the general (though jurisdictionally uneven) equivalence of withholding and withdrawing treatment (Beauchamp and Childress, already cited in Paper 2). The clause is narrower than the doctrine: it executes only toward less-burden states, time-boxed, with contemporaneous human confirmation where an underway intervention would be withdrawn, and automatic reversion absent confirmation. It does not adjudicate the jurisdictions where withdrawal is legally distinct from withholding;

module dossiers register those constraints per deployment (Paper 10), and the clause is disabled where they are not registered.

What this paper does not do. It does not decide when a patient should move between states; certification belongs to authorized humans, and the engine executes transitions it does not originate. It does not provide the utility functions within states; those are domain content, priced by evidence per Paper 5 and amended per the governance path. It does not claim the lattice's states are the right partition of the clinical continuum; they are the program's registered choice, falsifiable through the death-to-learning pipeline of §8. It does not resolve the ethics of goals-of-care conversations; it mandates that the conversation happen and be recorded, which is less than resolution and more than the status quo SUPPORT documented. And it does not demonstrate clinical benefit: H43–H49 are stated, and none is tested at seal time.

Appendix H-E — O-2 Restated (replaces the O-2 statement of §5.3)

O-2 — NO RAPID ALTERNATION UNDER BOUNDED-AMPLITUDE OSCILLATION

Let e(t) be the operative evidence strength, θ_descend the descent threshold, and δh > 0 the hysteresis margin of Eq. 5-1, and let W > 0 be a minimum dwell time registered per domain (governance parameter, §6.3's sibling). If the evidence oscillation is amplitude-bounded around the threshold — sup_t |e(t) − θ_descend| < δh over the window — then the operative objective admits no rapid alternation: any two consecutive transitions are separated by at least W, and no reversal (ascent after descent, or descent after ascent) can occur while the excursion remains within the band. The mechanism is the standard hysteresis argument with a time bound attached: each descent raises the ascent threshold to θ_descend + δh, which a sub-δh excursion cannot reach, and each ascent restores the descent threshold symmetrically, so within the band at most one direction is reachable at a time, and the dwell predicate forbids even that one from re-firing inside W.

This restatement corrects the earlier formulation, which claimed the objective "changes at most once" under sub-δh oscillation. That claim is false as stated: a slow drift within the band, or an excursion that crosses the band's edge after a large time gap, can license further transitions, and nothing in the thresholds alone bounds their total count. What hysteresis actually buys is the exclusion of rapid alternation — thrash — and O-2 now says exactly that and no more.

FORMALIZED-HEREIN — obligation, as a stability property over the transition calculus. Killed by: a logged deployment sequence containing two opposite-direction transitions with inter-transition time below W and inter-transition evidence deltas below δh (H46, §10).